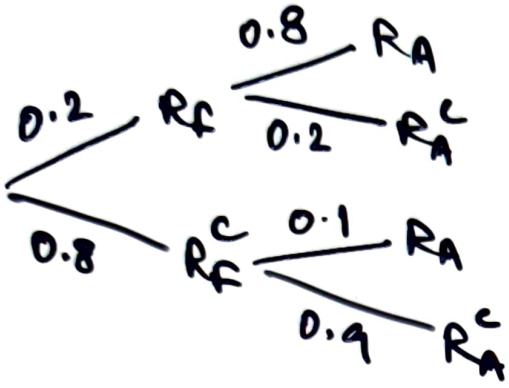
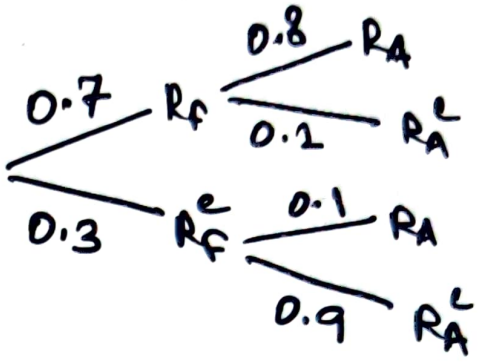
Problem set 2: MIT 6.041 Probability Systems Analysis

1. Let R_f = forecast that it rains

R_A = It actually rains



Summer and spring



fall and winter

1(a) Summer: $P(R_A) = P(R_A|R_f)\,P(R_f) + P(R_A|R_f^c)\,P(R_f^c)$

$= 0.2 \times 0.8 + 0.8 \times 0.1 = 0.24$

$$P(R_f|R_A) = \frac{P(R_A|R_f)\,P(R_f)}{P(R_A)} = \frac{2}{3}$$

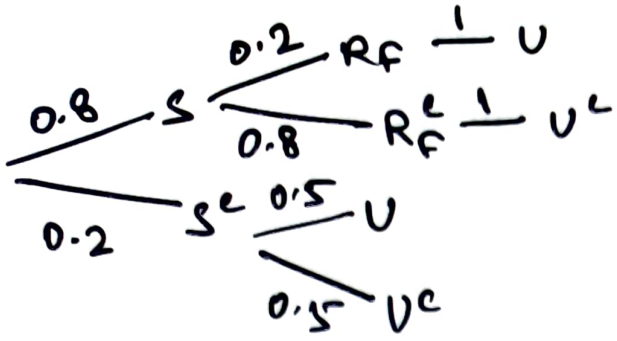
Winter: $P(R_A) = P(R_A|R_f)\,P(R_f) + P(R_A|R_f^c)\,P(R_f^c)$

$= 0.7 \times 0.8 + 0.3 \times 0.1 = 0.59$

$$P(R_f|R_A) = \frac{P(R_A|R_f)\,P(R_f)}{P(R_A)} = \frac{56}{59}$$

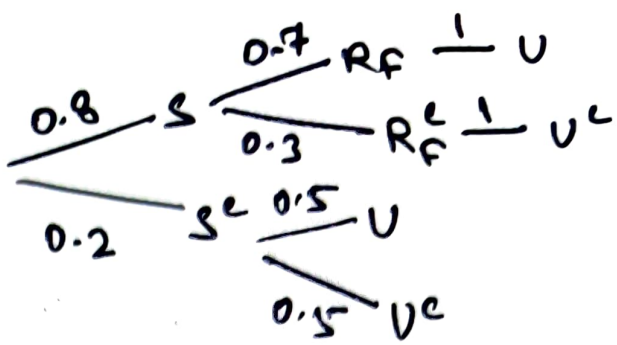
(b) Let S = Victor sees the forecast

U = Victor carries the umbrella



Summer and spring

$P(R_F^c) = 0.8$



Fall and winter

$P(R_F^c) = 0.3$

$$P(U) = P(U \mid R_F \cap S)\, P(R_F \mid S) P(S) + P(U \mid S^c) P(S^c) + P(U \mid R_F^c \cap S)\, P(R_F^c \mid S) P(S)$$

Summer and spring

$P(U) = 0.2 \times 0.8 + 0.2 \times 0.5$

$= 0.26$

Winter and fall

$P(U) = 0.7 \times 0.8 + 0.2 \times 0.5$

$= 0.66$

$$P(R_F^c \cap U) = P(R_F^c \cap U \cap S) + P(R_F^c \cap U \cap S^c)$$

$$= P(U \mid R_F^c \cap S) P(R_F^c \mid S) P(S) + P(U \mid R_F^c \cap S^c) P(R_F^c \mid S^c) P(S^c)$$

Summer and spring

$P(R_F^c \cap U) = 0.2 \times 0.8 \times 0.5$

$= 0.08$

$\neq P(R_F^c) P(U)$

Winter and fall

$P(R_F^c \cap U) = 0.2 \times 0.3 \times 0.5$

$= 0.03$

$\neq P(R_F^c) P(U)$

Given the season, R_F^c and U are not conditionally independent.

(c) $P(U \cap R_A^c) = P(U|R_A^c \cap R_F^c \cap S) P(R_A^c \cap R_F^c \cap S) +$

$P(U|R_A^c \cap R_F \cap S) P(R_A^c \cap R_F \cap S) +$

$P(U|R_A^c \cap R_F^c \cap S^c) P(R_A^c \cap R_F^c \cap S^c) +$

$P(U|R_A^c \cap R_F \cap S^c) P(R_A^c \cap R_F \cap S^c)$

$P(U|R_A^c \cap R_F^c \cap S) = P(U|R_F^c \cap S) = 0$

$P(U|R_A^c \cap R_F \cap S) = P(U|R_F \cap S) = 1$

$P(U|R_A^c \cap R_F^c \cap S^c) = P(U|R_A^c \cap R_F \cap S^c) = 0.5 = P(U|S^c)$

$\Rightarrow P(U \cap R_A^c) = P(R_A^c \cap R_F \cap S) + 0.5 P(R_A^c \cap R_F^c \cap S^c)$

$+ 0.5 P(R_A^c \cap R_F \cap S^c)$ ——— (1)

$P(S \cap U \cap R_A^c) = P(S \cap R_F \cap U \cap R_A^c) +$

$P(S \cap R_F^c \cap U \cap R_A^c)$

$= P(S \cap R_F \cap U \cap R_A^c)$

$= P(U|R_A^c \cap R_F \cap S) P(R_A^c \cap R_F \cap S)$

$= P(R_A^c \cap R_F \cap S)$ ——— (2)

$\Rightarrow P(S|U \cap R_A^c) = \dfrac{P(S \cap U \cap R_A^c)}{P(U \cap R_A^c)} = \dfrac{P(R_A^c \cap R_F \cap S)}{P(U \cap R_A^c)}$

——— (3)

① : $P(R_A^c \cap R_F \cap S) = P(R_A^c \mid R_F \cap S) P(R_F \mid S) P(S)$

$= P(R_A^c \mid R_F) P(R_F \mid S) P(S)$

$= P(R_A^c \mid R_F) P(R_F) P(S)$

$P(R_A^c \cap R_F^c \cap S^c) = P(R_A^c \mid R_F^c) P(R_F^c \mid S^c) P(S^c)$

$= P(R_A^c \mid R_F^c) P(R_F^c) P(S^c)$

$P(R_A^c \cap R_F \cap S^c) = P(R_A^c \mid R_F) P(R_F \mid S^c) P(S^c)$

$= P(R_A^c \mid R_F) P(R_F) P(S^c)$

$\Rightarrow P(U \cap R_A^c) = P(R_A^c \mid R_F) P(R_F) P(S) +$

$0.5 (P(R_A^c \mid R_F^c) P(R_F^c) P(S^c) + P(R_A^c \mid R_F) P(R_F) P(S^c))$

summer and spring :

$P(U \cap R_A^c) = 0.2 \times 0.2 \times 0.8 + 0.5 (0.8 \times 0.9 \times 0.2 +$

$0.2 \times 0.2 \times 0.2) = 0.108$

$P(R_A^c \cap R_F \cap S) = 0.2 \times 0.2 \times 0.8 = 0.032$

winter and fall :

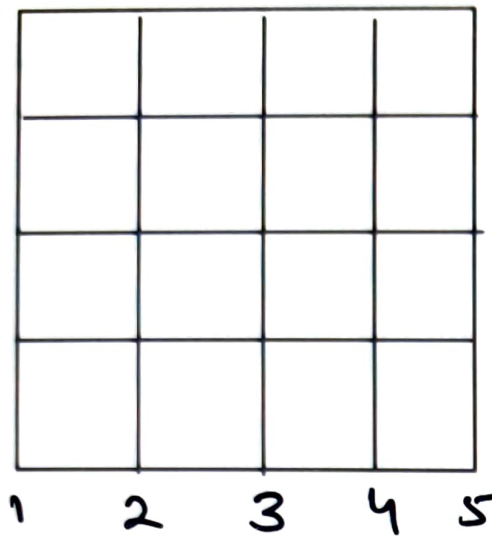
$P(U \cap R_A^c) = 0.2 \times 0.7 \times 0.8 + 0.5 (0.3 \times 0.9 \times 0.2 +$

$0.2 \times 0.7 \times 0.2) = 0.153$

$P(R_A^c \cap R_F \cap S) = 0.2 \times 0.7 \times 0.8 = 0.112$

$P(S \mid U \cap R_A^c) = \begin{cases} 8/27 \rightarrow \text{summer and spring} \\ 112/153 \rightarrow \text{winter and fall} \end{cases}$

2.



Let s_1 = outcome of roll 1
s_2 = outcome of roll 2

(a) $A = \{(s_1, s_2) \mid s_1 + s_2 = 10\}$

$B = \{(s_1, s_2) \mid s_1 = 5 \text{ or } s_2 = 5\}$

$C = \{(s_1, s_2) \mid s_1 = 1 \text{ or } s_2 = 1\}$

i) $B = \{(1,5), (2,5), (3,5), (4,5), (5,5), (5,1), (5,2), (5,3), (5,4)\}$

$A = A \cap B = \{(5,5)\} \subset B$

$\Rightarrow$ A and B are not independent

ii) $C = \{(1,1), (1,2), (1,3), (1,4), (1,5), (2,1), (3,1), (4,1), (5,1)\}$

$A = \{(5,5)\}$. $A \cap C = \phi$

$\Rightarrow$ A and C are not independent

(b) $D = \{(s_1, s_2) \mid s_1 + s_2 = 7\}$

$= \{(2,5), (3,4), (4,3), (5,2)\}$

$E = \{(s_1, s_2) \mid |s_1 - s_2| = 1\}$

$= \{(2,3), (3,2), (3,4), (4,3), (4,5), (5,4)\}$

$F = \{(s_1, s_2) \mid s_2 > s_1\}$

$\Rightarrow F = \{(1,2), (1,3), (1,4), (1,5), (2,3), (2,4), (2,5), (3,4), (3,5), (4,5)\}$

i) $E \cap F = \{(2,3), (3,4), (4,5)\} \Rightarrow P(E \cap F) = 3/25$

$P(E)P(F) = 6/25 \times 10/25 = 12/125 \neq P(E \cap F)$

$\Rightarrow$ E and f are not independent

ii) $E \cap F \cap D = \{(3,4)\} \Rightarrow P(E \cap F \cap D) = 1/125$

$$P(E \cap F | D) = \frac{P(E \cap F \cap D)}{P(D)} = 1/4$$

$E \cap D = \{(3,4), (4,3)\} \Rightarrow P(E \cap D) = 2/125$

$$P(E|D) = \frac{P(E \cap D)}{P(D)} = 1/2$$

$F \cap D = \{(2,5), (3,4)\} \Rightarrow P(F \cap D) = 2/125$

$$P(F|D) = \frac{P(F \cap D)}{P(D)} = 1/2$$

$P(E \cap F | D) = P(E|D) P(F|D)$

$\Rightarrow$ E and f are independent given D.

3. # old = 500 # defective and old = 75

new = 1500 # defective and new = 75

total = 2000 # total defective = 150

(a) Coin toss → two objects of same type

$\Rightarrow P(\text{old}) = P(\text{new}) = 0.5$

D_1 = item 1 is defective D_2 = item 2 is defective

$$P(D_1D_2) = P(D_1D_2 \mid \text{old})\,P(\text{old}) + P(D_1D_2 \mid \text{new})\,P(\text{new})$$

$$P(D_1D_2 \mid \text{old}) = P(D_1 \mid \text{old})\,P(D_2 \mid D_1 \cap \text{old})$$

$$= \frac{75}{500} \times \frac{74}{499} = 0.0222$$

$$P(D_1D_2 \mid \text{new}) = P(D_1 \mid \text{new})\,P(D_2 \mid D_1 \cap \text{new})$$

$$= \frac{75}{1500} \times \frac{74}{1499} = 0.0025$$

$$\Rightarrow P(D_1D_2) = 0.5 \times (0.0222 + 0.0025) = 0.01235$$

$$(b)\ P(\text{both old} \mid D_1D_2) = \frac{P(D_1D_2 \mid \text{old})\,P(\text{old})}{P(D_1D_2)}$$

$$= 0.8988$$

4. $P(\text{dog in A}) = 0.4$, $P(\text{dog in B}) = 0.6$

$P(\text{search in A} \mid \text{dog in A}) = 0.25$

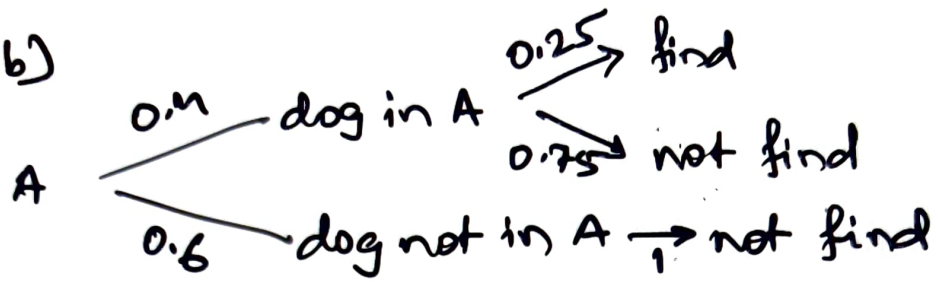
$P(\text{search in B} \mid \text{dog in B}) = 0.15$

(a) $P(\text{search in A} \cap \text{dog in A}) = 0.4 \times 0.25 = 0.1$

$P(\text{search in B} \cap \text{dog in B}) = 0.6 \times 0.15 = 0.09$

⇒ he should search in forest A.

(b)



$P(\text{not find}) = P(\text{not find} \cap \text{dog in A}) + P(\text{not find} \cap \text{dog not in A})$

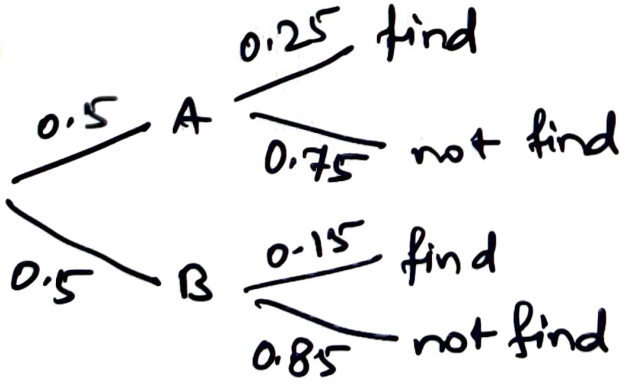
$= P(\text{not find} \mid \text{dog in A})\, P(\text{dog in A}) + P(\text{not find} \mid \text{dog not in A})\, P(\text{dog not in A})$

$= 0.75 \times 0.4 + 0.6 = 0.9$

$P(\text{dog in A} \mid \text{not find}) = \dfrac{P(\text{not find} \cap \text{dog in A})}{P(\text{not find})}$

$= \dfrac{0.3}{0.9} = \dfrac{1}{3}$

(c)



$P(\text{find} \cap A) = P(\text{find} \mid A)\, P(A) = 0.125$

$P(\text{find} \cap B) = 0.075$

$P(\text{search in A} \mid \text{find}) = \dfrac{P(\text{find} \cap A)}{P(\text{find})} = \dfrac{5}{8}$

(d) $P(\text{die on day } N) = \frac{N}{N+2}$

$P(\text{finds in A alive on day 2})$

$= P(\text{does not find on day 1} \cap \text{looks in A})$

$P(\text{finds on day 2} \cap \text{looks in A})\ P(\text{alive on day 1})\ P(\text{alive on day 2})\ P(\text{in A})$

$= 0.75 \times \frac{2}{3} \times 0.25 \times \frac{1}{2} \times 0.4 = 0.025$

5. (a) A, B, and C are independent

$P(A \cap B) = P(A)P(B)$, $P(B \cap C) = P(B)P(C)$,

$P(A \cap C) = P(A)P(C)$ and

$P(A \cap B \cap C) = P(A)P(B)P(C)$

$P(A \cap (B \cup C)) = P((A \cap B) \cup (A \cap C))$

$= P(A \cap B) + P(A \cap C) - P(A \cap B \cap C)$

$= P(A)P(B) + P(A)P(C) - P(A)P(B)P(C)$

$= P(A)(P(B) + P(C) - P(B)P(C))$

$= P(A)(P(B) + P(C) - P(B \cap C))$

$= P(A)P(B \cup C)$

$\Rightarrow$ A and B $\cup$ C are independent

(b) $A_1, A_2, \ldots, A_n$ are independent events

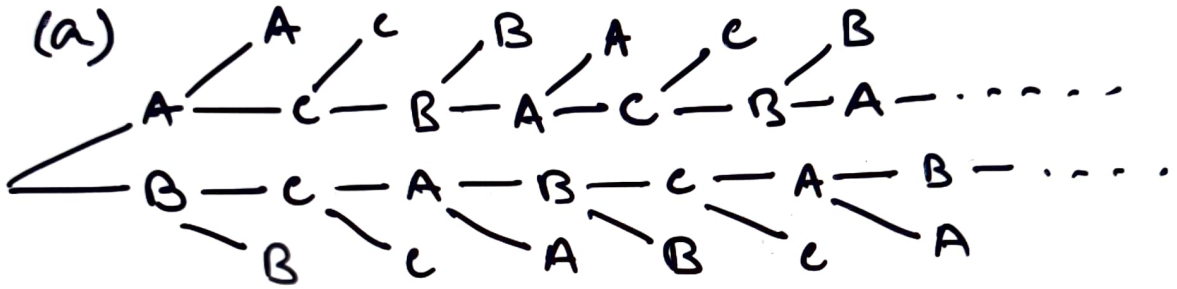
$\Rightarrow A_1^c, A_2^c, \ldots, A_n^c$ are independent events

$$A_1 \cup A_2 \cup \ldots \cup A_n = (A_1^c \cap A_2^c \cap \ldots \cap A_n^c)^c$$

$$\Rightarrow P(A_1 \cup A_2 \cup \ldots A_n) = 1 - P(A_1^c \cap A_2^c \cap \ldots \cap A_n^c)$$

$$= 1 - \prod_{i=1}^{n} (1 - P(A_i))$$

G1. (a)



(b) minimum # games to finish tournament = 2

maximum # games to finish tournament = ∞

2 games : {AA, BB}, 3 games : {ACC, BCC},

4 games : {ACBB, BCAA}, ...

every k game history has two possible options.

P(tournament ends in k games) =

ways tournament ends in k games $\times$ P(k)

$$= 2 \times \frac{1}{2^k} = \frac{1}{2^{k-1}}$$

$$\Rightarrow \sum_{k=2}^{\infty} P(\text{tournament ends in } k \text{ games})$$

$$= \frac{1}{2} + \frac{1}{2^2} + \ldots \infty = 1$$

⇒ This is a valid probability law.

(c) $\{\text{At most 5 games}\} = \{\text{At least 6 games}\}^c$

$P(\{\text{At most 5 games}\}) = 1 - P(\{\text{At least 6 games}\})$

$P(\{\text{At least 6 games}\}) = \sum_{k=6}^{\infty} \frac{1}{2^{k-1}} = \frac{1}{16}$

⇒ $P(\{\text{Tournament ends in at most 5 games}\})$

$= 1 - \frac{1}{16} = \frac{15}{16}.$

Alice wins :

At most 5 games : $\{AA, ACBAA, BCAA\}$

⇒ $P(\text{Alice wins in at most 5 games}) = 1/4 + 1/16 + 1/32$

$= 11/32$

∞ games : $\{AA, BCAA, ACBAA, BCABCAA, ACBACBAA, \ldots\}$

⇒ $P(\text{Alice wins}) = 1/2^2 + 1/2^4 + 1/2^5 + 1/2^7 + 1/2^8 + \cdots$

$= (1/2^2 + 1/2^5 + 1/2^8 + \cdots) + (1/2^4 + 1/2^7 + \cdots)$

$= 1/4 \cdot 8/7 + 1/16 \cdot 8/7 = 2/7 + 1/14 = 5/14.$

Bob wins :

At most 5 games : $\{BB, ACBB, BCABB\}$

$\Rightarrow$ P(Bob wins in at most 5 games) = $1/4 + 1/16 + 1/32$

$= 11/32$

∞ games : { BB, ACBB, BCABB, ACBACBB, BCABCABB, ... }

$\Rightarrow$ P(Bob wins) = $1/2^2 + 1/2^4 + 1/2^5 + 1/2^7 + 1/2^8 + \cdots$

$= (1/2^2 + 1/2^5 + 1/2^8 + \cdots) + (1/2^4 + 1/2^7 + \cdots)$

$= 1/4 \cdot 8/7 + 1/16 \cdot 8/7 = 2/7 + 1/14 = 5/14$.

Carol wins:

At most 5 games : { ACC, BCC }

$\Rightarrow$ P(Carol wins in at most 5 games) = $1/8 + 1/8 = 1/4$

∞ games : { ACC, BCC, ACBACC, BCABCC, ... }

$\Rightarrow$ P(Carol wins) = $1/2^3 + 1/2^3 + 1/2^6 + 1/2^6 + \cdots$

$= 2 \times (1/2^3 + 1/2^6 + \cdots) = 2 \times 1/8 \times 8/7 = 2/7$

$= 1 - P(\text{Alice wins}) - P(\text{Bob wins})$